Sri Harsha Pillalamarri

Systems Engineer | AI Engineer | Generative AI

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Professional Summary

AI Engineer with 2+ years of experience at Tata Consultancy Services (TCS) building enterprise Generative AI solutions for North American BFSI clients. Experienced in designing agentic AI workflows using LangChain and LangGraph for legacy application modernization, code migration, business rule extraction and automated validation. Hands-on experience with RAG, prompt engineering, embeddings, LLM evaluation, Amazon Bedrock, FastAPI and AWS cloud services.

Technical Skills

Programming Languages: Python, C#, SQL, Java
Generative AI: LangChain, LangGraph, Prompt Engineering, Agentic AI, Deep Agents, Retrieval Augmented Generation (RAG), Sentence Transformers, Embeddings,LLM Evaluation
LLMs: Amazon Bedrock Models, Claude, HuggingFace Models
AI Engineering: Multi-Agent Systems, Code Generation, Code Modernization, Business Rule Extraction, Code Parsing, Golden Dataset Evaluation, Regression Evaluation
Backend: FastAPI, REST APIs, Async Python
Cloud: AWS EC2, ECS, Bedrock, SageMaker, S3, CloudWatch
Databases: Oracle, SQL Server, Snowflake
Machine Learning / Computer Vision: YOLOv5, Object Detection, Deep Learning, Computer Vision
Frameworks & Tools: .NET, Angular, Git, AppDynamics

Professional Experience

Tata Consultancy Services (TCS) June 2024 – Present
Systems Engineer | Generative AI Engineer
BFSI-Notth America CTO Team

- Designed and implemented LangGraph-based multi-agent orchestration for Legacy Code Modernizations. Currently working for Hartford insurance.
- Integrated foundation models through Amazon Bedrock using LangChain-based orchestration pipelines for AI-assisted code conversion and modernization.
- Worked on multiple legacy modernization demonstrations involving Delphi, PowerBuilder, Oracle Forms, C#/.NET and other legacy application technologies, with modernization targets including Java and modern application architectures.
- Created golden evaluation datasets containing reference modernization outputs to validate AI-generated code and measure the quality and correctness of modernization workflows.
- Contributed to enterprise modernization initiatives including .NET version upgrades, Oracle Forms business rule extraction and AI-assisted C# to Angular migration.
- Worked with FastAPI and asynchronous Python to expose AI modernization capabilities through backend APIs and integrate LLM workflows with enterprise applications.
- Worked with AWS EC2, ECS, SageMaker, Bedrock, CloudWatch and S3 for deployment, experimentation, monitoring and troubleshooting of AI services.

Projects

Customer Support RAG Assistant
Python | LangChain | RAG | Sentence Transformers | FastAPI

- Built a Retrieval-Augmented Generation (RAG) based customer support assistant for answering queries from support documentation and knowledge-base content.
- Implemented document ingestion, preprocessing, chunking and Sentence Transformer embeddings to create a searchable knowledge base.
- Developed semantic retrieval to identify relevant support articles, troubleshooting information, product documentation and frequently asked questions for user queries.
- Exposed the RAG pipeline through FastAPI REST APIs for integration with customer-support applications.

Traffic Detection and Optimization
Python | YOLOv5 | Computer Vision | Deep Learning

- Developed a computer-vision based traffic monitoring and optimization system using the YOLOv5 object detection algorithm.
- Processed traffic video and image data to detect and classify vehicles and analyze vehicle density across road intersections.
- Applied object-detection results to estimate traffic conditions and support dynamic traffic-flow optimization.
- Worked with deep-learning based object detection, model inference, data preprocessing and evaluation of detection results.
- Analyzed vehicle counts and traffic patterns to support more efficient allocation of traffic signal timing.

Certifications

- Claude Certified Architect
- AWS Certified Data Engineer – Associate
- Google Cloud Computing Foundations Certificate on Google Skills

Education

Bachelor of Technology in Computer Science and Engineering (Data Science)
Gayatri Vidya Parishad College of Engineering
CGPA: 8.8